

PBSMS

Tenant AI Assistant

SRS v2.1 · Chapter 47 · Version 2.1 · Adopted Baseline · Complete consolidated text

Item	Value
Document	PBSMS Tenant AI Assistant — SRS v2.1, Chapter 47
Version	2.1 — Adopted Baseline. Complete consolidated text. Supersedes v2.0 and retires the v2.1 amendment schedule.
Status	ADOPTED BASELINE — APPROVED FOR BUILD. Adopted 29 August 2026. Build authorisation is staged; see §47.0.2.
Reading note	This is the whole chapter. It is not a diff. Nothing needs to be read alongside it to know what is required.
Requirement prefixes	FR-AIT- (functional) · NFR-AIT- (non-functional) · TEN-05x/06x (tenancy) · DP-1xx (data protection)
Governs	A named, versioned PBSMS Domain AI Model and its mandatory governance infrastructure, delivering capabilities within each requesting user's effective scope
Depends on	Ch 13 (access scope) · Ch 23 (finance model) · Ch 34 (frontend) · Ch 40 (retention) · FR-ANL-040
Implementation baseline	PBSMS Domain AI Model Architecture and Implementation Blueprint v1.0, adopted 29 August 2026. Subordinate to this chapter; see §47.0.4.
Companion	Internal Engineering Agent v2.0 (internal process, not an SRS chapter). An amendment to that document is outstanding; see §47.0.5.

Revision Note — v2.0 → v2.1

v2.1 makes a named, versioned PBSMS Domain AI Model the primary artifact, and adds four capabilities the v2.0 text did not permit: role-scoped Research, Controlled Live Internet Research, General Conversation, and Combined Analysis. Because v2.0 states that anything not expressly permitted is prohibited, each had to be specified rather than inferred.

This version is issued as complete text rather than as an amendment schedule. A requirement is read many times and amended rarely; a reader should never have to reconstruct the current rule from a baseline plus a patch list. The amendment schedule dated 28 August 2026 and its gap-closure schedule are retained as the review record and are not build documents.

#	Change	Reason
1	Model-first framing throughout (§47.1, new FR-AIT-700 to 705)	The model is the intelligence being built. Retrieval, validation and audit are its enforcement boundary, not a substitute for it.
2	New capabilities: Research, Live Internet Research, General Conversation, Combined Analysis	Explicitly permitted with their own isolation, privacy and citation rules.

#	Change	Reason
3	FR-AIT identifiers organised into blocks by subject	Sequential continuation placed research and injection requirements inside the audit block. IDs are permanent once adopted, so this could only be fixed before adoption.
4	New FR-AIT-904, 905 and DP-118 — direct-input confidential content	General Conversation creates an ingress path that bypasses retrieval, so DP-100 does not reach it.
5	FR-AIT-903 replaces mode-based session termination with evidence-based	Terminating a lesson-planning conversation on a campus switch carried a usability cost and no isolation benefit.
6	New FR-AIT-706, 707 — validation block rate is measured and gated	Every other blocking gate measures displayed responses. A poorly grounded model passes them and the cost surfaces as blocked responses nothing was watching.
7	Build order rebuilt around model development, with a demand-validation phase first	The governance runtime is required under every outcome. The model programme is contingent and expensive, so the evidence justifying it comes first.

47.0 Approval and Change Control

This chapter is adopted into the SRS v2.1 baseline as of 29 August 2026 and is approved for build. It carries the same standing as any other chapter of the SRS: it is binding on implementation, and where an implementation and this chapter disagree, this chapter wins until it is formally amended. Chapter 47 v2.0 is archived and is no longer a build document.

47.0.1 Amendment process

- Any change to a requirement requires a version increment and re-adoption. Requirement IDs are permanent — a withdrawn requirement is marked withdrawn and retained, never deleted and never reused.
- Amendments are issued as complete replacement text, not as schedules of edits against a prior version.
- Any implementation departing from the literal text here shall be recorded as a named deviation with its reasoning, on the pattern of the Frontend Design Specification §14, and approved by the Engineering Lead before it ships. A silent deviation is a defect regardless of whether the resulting behaviour is better.
- The blocking thresholds in §47.16.2 are not amendable downward. They may be raised. Lowering any of them requires re-adoption of this chapter as a whole, because their function is to be inconvenient at the moment someone wants to move them.

47.0.2 Build authorisation

Approval authorises the build. It does not clear every stage, because several prerequisites are external to engineering and cannot be closed by starting work. Stages are defined in §47.17.

Stage	Scope	Authorisation
0.5	Demand validation — Ask only, staff only, read-only, on the governance runtime with a prompt-configured general model	CLEARED. Internal and consenting-canary measurement only. Never released, marketed or described as the PBSMS Domain AI Model (FR-AIT-

Stage	Scope	Authorisation
	behind the model adapter	702).
1-2	Model definition, registry, constitution; synthetic corpus, golden sets, adversarial corpus, evaluation harness	CLEARED. No tenant data, no production credentials, no unrestricted internet.
3	Base-model selection and PBSMS adaptation	CLEARED for work on approved non-tenant data only.
4-5	Governance runtime; operational retrieval under RLS	CLEARED. Required under every outcome, including abandonment of the model programme.
6	Research collections and Controlled Live Internet Research	GATED on the privacy, DLP and network gates in §47.8 and §47.16.2 being green, and on search and browsing providers assessed under DP-113.
7	Live tenant retrieval — operational shadow	GATED on a model-hosting arrangement contractually satisfying DP-103, DP-104 and DP-107.
8-9	Canary, then Draft, Explain, Find, Combined Analysis, demo and metering	GATED on the preceding stage. Commercial limits in §47.11 required before demo and metering.
10	Parent View	NOT AUTHORISED by this adoption. Separate decision and separate risk review.
—	FR-AIT-060 multilingual capability	NOT AUTHORISED until evaluation sets exist for Twi, Ga, Ewe and Hausa in this domain.

Stages 0.5 through 5 are cleared together and deliberately. They are the demand evidence, the isolation boundary, the evaluation harness and the enforcement runtime — the parts that either cannot be retrofitted or are required whatever model eventually sits behind them. None depends on the provider decision that gates live tenant access.

47.0.3 What this approval does not settle

The open questions in §47.19 survive adoption. Adoption means the requirements are binding, not that every unknown is closed — a baseline that waited for that would never be adopted.

47.0.4 Relationship to the Architecture Blueprint

The PBSMS Domain AI Model Architecture and Implementation Blueprint v1.0 is adopted alongside this chapter as the implementation baseline. It describes how the requirements here are to be built: reference architecture, trust and network zones, model profile, constitution compilation, tool and evidence contracts, the research gateway lifecycle, dataset families, training stages, serving runtime, storage design, deployment and rollback.

The two documents are not peers.

- This chapter states what is required. The Blueprint states how to build it. Where they conflict, this chapter controls, and the Blueprint is corrected — not the reverse.
- The Blueprint may not lower a threshold, relax a prohibition, authorise a surface, or widen a scope defined here. Its own text already says so; this records it from the controlling side.

- An amendment to the Blueprint does not amend this chapter. Where implementation experience shows a requirement here is wrong, the correction is a version increment and re-adoption of this chapter under §47.0.1, not a Blueprint revision that quietly diverges from it.
- Blueprint values presented as reference decisions — the 8B to 14B candidate envelope, the 32k context floor, the initial corpus distribution, the reference training stack — are starting points for measurement, not adopted requirements. They carry no normative force and are expected to move.

One consequence of adopting the Blueprint is immediate and outside this chapter. Blueprint §14.2 establishes the pbsms-ai/ ownership boundary as the working repository layout. That tree holds the Model Constitution, the dataset manifests, and the blind evaluation corpus on which every release decision under §47.16 depends. §47.0.5 records what follows.

47.0.5 Outstanding companion amendment

Internal Engineering Agent v2.0 prohibits the Agent from modifying "the Assistant's prompts, scope configuration, retrieval rules or evaluation gates" (EC-005), and makes four named test suites immutable to it (EC-400). Both lists were written before this chapter had model artefacts or evaluation corpora, and neither reaches the pbsms-ai/ tree.

As those documents currently stand, the Engineering Agent may modify the blind evaluation corpus. A modification there would fail no CI gate, because the gates are computed against that corpus, and it would silently invalidate every certification downstream of it. The Agent must be amended to place the whole pbsms-ai/ boundary out of write reach, with read access for question answering retained.

This is recorded here because adopting the Blueprint is what makes the exposure live. It is not a defect in this chapter and does not block anything in it — but it should be closed before any work begins inside that tree, which under §47.18 is stage 1.

47.1 Purpose and Position in the System

The Tenant AI Assistant is delivered by the PBSMS Domain AI Model: a named, versioned, domain-adapted model developed for school administration, education support, scoped research and general assistance. The model is the primary artifact. It interprets requests, plans permitted retrieval, uses authorised tools, composes grounded answers and drafts, identifies ambiguity, and produces structured refusals.

The model operates inside mandatory governance infrastructure. That infrastructure authenticates the requester, computes effective scope, retrieves the minimum permitted evidence, controls tool and internet access, validates output, inserts exact values, applies human-approval gates, records audit evidence, meters cost, and blocks prohibited behaviour. The infrastructure governs and enforces the model; it does not replace it as the intelligence being developed.

The Assistant remains assistive. It is not a system of record, an authority, an autonomous agent, or a decision-maker. It holds no independent access entitlement. Its capabilities become available only through the permissions and context supplied for the current request.

A rule is never considered enforced because it appears in training data or a system prompt. Rules suitable for learning are taught. Every critical rule additionally has an independent runtime control, or a documented reason why deterministic enforcement is not technically possible.

47.1.1 Why the prohibitions carry equal weight

This product holds the personal data of minors, the financial records of fee-paying households, and academic results that determine BECE candidacy and CSSPS placement. An Assistant that is merely useful is not a good outcome; one that is useful and cannot be made to say something false about a child's money, attendance or results is the only acceptable one. Every capability below is specified with its failure mode named, and §47.16 treats groundedness, isolation, privacy and numeric fidelity as pass/fail release blockers rather than quality metrics to improve over time.

47.1.2 Defined terms

Term	Definition
PBSMS Domain AI Model	The named, versioned model artifact developed or adapted for PBSMS, including its checkpoint or adapter, tokenizer and configuration, Model Constitution, tool schemas, output schemas, model card, evaluation report and release manifest.
Model Constitution	The versioned set of behavioural rules defining permitted capabilities, prohibited behaviours, source boundaries, refusal conditions, tool-use rules and accountability requirements.
Operational Evidence	Live PBSMS records retrieved under the requesting user's effective scope. The only permitted source for tenant-specific facts.
Research Evidence	Material retrieved from an access-controlled research collection or through Controlled Live Internet Research, with claim-level citations and retrieval time.
General Explanation	Content drawn from the model's general reasoning and parametric knowledge. It may not assert a tenant fact or a current external fact.
Combined Analysis	A response using more than one source class while preserving separate permissions, retrieval paths, claim labels and citations.
Controlled Live Internet Research	Read-only, mediated internet access through a policy and privacy gateway. A model capability, not network access.
Effective Scope	The intersection of every applicable identity, tenancy, role, context, data, source, conversation, subscription and tool permission at request time.
Directly Supplied Content	Text a user types or pastes into a conversation. It is not Operational Evidence and never becomes it.

47.2 Scope

In scope	Out of scope
Natural-language questions over authorised PBSMS records	Any tenant, school, role, field, record, research source, tool or conversation outside effective scope
Role-scoped research over approved collections	Treating a research collection as visible merely because it belongs to the same tenant
Controlled Live Internet Research with citations	Unrestricted browsing, private-network access, autonomous posting, transactions, account creation or any external side effect
General conversation for education, administration,	Using general model knowledge to assert a tenant-

In scope	Out of scope
writing, explanation and planning	specific fact or an unverified current fact
Combined analysis over separately authorised operational and research evidence	Sending identifiable or confidential PBSMS data to a search engine, website or external research provider
Human-reviewed drafting in the existing approved artefact types	Sending, publishing, saving-as-final or modifying any PBSMS record without the existing human gate
Private conversations and explicitly authorised shared workspaces	Access to another user's private conversation or unshared research material
Model adaptation on approved synthetic, public, licensed or internally authored non-tenant data	Training, fine-tuning, embedding adaptation or preference learning from tenant records, uploads or conversations

47.3 Capabilities

Seven modes. Each is defined by what it returns and which source classes it may use.

Mode	Permitted knowledge	Output	Principal block
Records	Operational Evidence in scope	Referenced answer	No evidence, no tenant claim
Draft	Authorised operational and permitted research evidence	AI-marked draft	Existing human approval required
Explain	Owning module's displayed value plus lineage	Narrative walk-through	Model may not recompute
Find	Authorised structured or semantic records	Record set with count	Records, not unsupported prose
Research	Approved collections and controlled live internet	Cited synthesis	No protected data outbound
General	Model parametric knowledge	Conversation or writing	Cannot assert a tenant or live fact
Combined	Separately authorised operational and research evidence	Claim-labelled synthesis	Source classes stay separate

47.3.1 Ask

- **FR-AIT-010** — The Assistant shall answer natural-language questions over the requesting tenant's records, restricted to the requesting user's permission scope as defined in Chapter 13.
- **FR-AIT-011** — Every factual claim shall carry a reference to the record or records it derives from, rendered as navigable links. A claim that cannot be traced to a retrieved record shall not be returned.
- **FR-AIT-012** — Responses returning a set of records shall state the size of the set and whether it was truncated. Silently omitting matching records is a defect.

Worked example — a headmaster asks which JHS 2 students are below 80% attendance this term:

You asked: JHS 2 students below 80% attendance, Term 2

4 students, of 61 enrolled in JHS 2. Retrieved 11:04, today.

Ama Mensah	JHS 2A	71%	32 of 45 days	→ open
Kofi Boateng	JHS 2A	76%	34 of 45 days	→ open
Yaa Owusu	JHS 2B	64%	29 of 45 days	→ open
Kwesi Antwi	JHS 2B	78%	35 of 45 days	→ open

Source: attendance register, Term 2, 2026/2027 · 45 marked days

[Open absence follow-up queue]

Note what the example does not do: it does not say who is "at risk", does not rank by concern, and does not suggest a cause. Those are judgements about children (FR-AIT-104).

47.3.2 Draft

- **FR-AIT-020** — The Assistant shall draft text for human review in these places only: guardian messages (Ch 26), report-card comments (FR-DOC-010), term summaries, absence follow-up notes (FR-ATT-040), and admission decision letters (FR-ADM-030).
- **FR-AIT-021** — Every draft shall enter the existing approval control for that artefact. The Assistant shall not send, publish or save-as-final any content.
- **FR-AIT-022** — Drafts shall be marked AI-drafted in the approval interface and the audit record, and shall remain visually distinguishable from human-authored content until approved.

47.3.3 Explain

- **FR-AIT-030** — The Assistant shall explain how a displayed figure was derived by walking the records that produced it — a fee balance to its invoices, allocations and payments; a term aggregate to its component assessments and weights.
- **FR-AIT-031** — The Assistant shall read the value computed by the owning module and render it unchanged. It shall not recompute, re-derive or reconcile any figure (FR-AIT-101).

47.3.4 Find

- **FR-AIT-040** — Natural-language search across students, guardians, staff, invoices, payments and messages within the user's scope, returning records rather than prose.

47.3.5 Session and context

- **FR-AIT-050** — Conversation context shall be retained within a single session, scoped to one user and one tenant.
- **FR-AIT-051** — A session holding Operational Evidence shall additionally be bound to one school, one campus and one academic year, and shall terminate on a change to any of them.
- **FR-AIT-052** — Sessions shall expire with the user's authenticated session and shall not be resumable across logins.
- **FR-AIT-903** — Session continuity is determined by evidence, not by mode. A conversation that has at any point contained Operational Evidence is bound to the context in which that evidence was retrieved, and FR-AIT-050 to 052 apply in full. A conversation containing only General Explanation or Research Evidence survives a change of school, campus, academic year or term. A conversation transitions permanently to the bound state on its first turn containing Operational Evidence, and never transitions back.

The one-way transition matters. A Combined Analysis turn inside a previously general conversation pulls operational evidence into that context, and the conversation must inherit the stricter rule from that point

on. Allowing it to relax again once the evidence scrolls out of the window would be the quiet reintroduction the original rule exists to prevent.

47.3.6 Language

- **FR-AIT-060** — English by default. Where a tenant enables it, the Assistant shall accept questions in Twi, Ga, Ewe or Hausa and answer in the language of the question.
- **FR-AIT-061** — Parent-facing text drafted in a language the approving staff member has not confirmed they read shall carry a warning at the approval step. An approver cannot take accountability for text they cannot read.

FR-AIT-060 shall not ship on assertion. No evaluation set exists for these languages in this domain, and translation quality in Ghanaian languages for school-administrative register is unverified. It is gated on §47.16 like every other capability.

47.3.7 Effective capability scope

- **FR-AIT-720** — The model's general capabilities shall not create access entitlement. Every capability, source, tool, conversation and output shall be restricted to the requesting user's effective scope.
- **FR-AIT-721** — Before each model call, PBSMS shall construct a server-side capability manifest containing only the capabilities, source classes, retrieval tools, research collections, internet permissions, output types and limits authorised for that request. The model shall not be able to add to or modify this manifest.
- **FR-AIT-722** — A tenant administrator shall be able to enable or disable Research, Controlled Live Internet Research, General Conversation and Combined Analysis globally or per role, taking effect immediately.

47.3.8 Research

- **FR-AIT-800** — The Assistant shall answer research questions using research sources authorised for the requesting user, with claim-level citations and retrieval timestamps.
- **FR-AIT-801** — Approved research collections may contain public, licensed, tenant-owned, role-owned or user-owned documents. Read, upload, curate, share and remove permissions shall be separately assignable.
- **FR-AIT-802** — Where Controlled Live Internet Research is enabled, the model may formulate a research plan, submit privacy-safe queries through the research gateway, open permitted sources, compare evidence and synthesise a cited response.
- **FR-AIT-803** — Internet access shall be read-only. The model shall not post content, send messages, create accounts, sign in, purchase, enrol, upload tenant content, execute remote actions, or cause any external side effect.
- **FR-AIT-804** — No outbound query, URL parameter, request body or external tool call shall contain a student name, guardian name, personal identifier, protected record, credential, unpublished result, financial transaction detail or other confidential tenant content.
- **FR-AIT-805** — Every claim derived from external research shall identify its source by title, publisher or domain, URL or durable identifier, publication date where available, and retrieval timestamp.
- **FR-AIT-806** — The Assistant shall prefer authoritative and primary sources, state material source limitations, and show relevant disagreement where credible sources conflict. It shall not silently select a convenient source.

- **FR-AIT-807** — If live research is unavailable, insufficient, stale for the question, or blocked by policy, the Assistant shall state that condition. It shall not replace missing research with an apparently current answer from model memory.

47.3.9 General Conversation

- **FR-AIT-900** — Where enabled, the Assistant shall support general conversation for education, school administration, explanation, writing, planning and other lawful assistance that requires no access to unauthorised PBSMS data.
- **FR-AIT-901** — General model knowledge may support non-tenant-specific assistance. It shall not be used to state a fact about a tenant, school, user, student, guardian, staff member, result, attendance record, invoice, payment, message or other PBSMS record.
- **FR-AIT-902** — A private conversation shall be visible only to its owner unless the owner explicitly shares it into an authorised workspace. Tenant administrators may view audit metadata as permitted by law and policy, but shall not receive blanket access to private conversation content.

Directly Supplied Content

DP-100 excludes health and discipline records from retrieval. That control governs what the system fetches. General Conversation does not fetch — a user types or pastes directly. A staff member can therefore introduce a discipline summary, an unpublished result or a family's payment history into a conversation, and it is then stored as transcript content. Before this mode existed there was no ingress path that bypassed retrieval.

- **FR-AIT-904** — Directly Supplied Content shall be treated as tenant-confidential by default. It shall not be transmitted to any external research destination, shall not be included in a parent-facing drafted artefact, and shall not become Operational or Research Evidence for any claim. It may inform explanation and drafting for the supplying user only.
- **FR-AIT-905** — Where Directly Supplied Content is detected as matching a restricted category under DP-100, or as containing personal identifiers, the interface shall state that the content will be retained in the transcript and shall offer to discard the turn. The Assistant shall not silently accept confidential content it would have refused to retrieve.

Honest limitation: detection of pasted confidential content is heuristic and will both miss cases and produce false positives. FR-AIT-904 is the load-bearing rule because it is deterministic — it governs the destination regardless of whether detection worked. FR-AIT-905 improves the outcome; it does not carry it.

47.3.10 Combined Analysis

- **FR-AIT-820** — Each response shall classify every factual claim as Operational Evidence, Research Evidence or General Explanation. The interface shall render the source class clearly.
- **FR-AIT-821** — Combined Analysis may use two or more source classes only where the requesting user is authorised for every source and tool involved. Retrieval channels and permission checks shall remain separate.
- **FR-AIT-822** — An official value or status from the owning PBSMS module shall take precedence over external research and general model knowledge. External material may explain context but shall not override, reconcile or silently correct a PBSMS record.
- **FR-AIT-823** — Where a response mixes source classes, each claim and citation shall remain attributable after export, copying or conversion into a permitted draft.

Permitted example: a headmaster asks the Assistant to review an authorised attendance summary and research current Ghanaian approaches to reducing JHS absenteeism. Authorised aggregates go to the model; the outbound query contains only a general formulation. The model may describe general interventions. It may not profile or predict outcomes for individual children.

47.4 Prohibited Behaviours

Requirements, not guidance. Each is testable and each is a blocking gate in §47.16.

- **FR-AIT-100** — No writes. The Assistant shall have no write path to any tenant record. It cannot mark attendance, enter or alter scores, publish or reopen results, create or void invoices, capture, allocate or reverse payments, alter permissions, or change configuration. Its output is text and record references only.
- **FR-AIT-101** — Not a second source of financial truth. The Assistant shall not compute, derive, estimate or restate any monetary figure. Every amount shall be read from the finance module's computed value and rendered unchanged. An AI-produced figure disagreeing with the ledger is a severity-one defect.
- **FR-AIT-102** — No unpublished results. Draft, in-review and returned results (FR-RES-010) shall be excluded from retrieval for parent and student users unconditionally. For staff users, any draft result appearing in a response shall be labelled as draft.
- **FR-AIT-103** — No determinations. The Assistant shall not generate a grade, promotion recommendation, admission outcome, disciplinary finding, or any other determination affecting a child's academic record or standing. It may summarise such a determination after a human has made it.
- **FR-AIT-104** — No profiling of children. No predictions, risk scores, behavioural characterisations or comparative judgements about an individual student. "Which students are below 80% attendance" is a query over recorded facts and is permitted. "Which students are likely to fail" is a prediction about a child and shall be refused with a stated reason.
- **FR-AIT-105** — No cross-tenant anything. No answer, aggregate, benchmark, comparison or example shall incorporate data from any tenant other than the requesting one, including anonymised or aggregated. Where a tenant asks how they compare to other schools, the Assistant shall state that the platform does not provide cross-school comparison.
- **FR-AIT-106** — No self-modification. No access to system configuration, feature flags, source code, schema, the Model Constitution, or its own prompts, and no capability to alter its own behaviour, scope or permissions.
- **FR-AIT-150** — Instructions contained in Operational Evidence, Research Evidence, a web page, document, search result, citation or metadata shall be treated as data and shall not alter the Model Constitution, capability manifest, system instructions, tool permissions or output rules.
- **FR-AIT-151** — The model shall not acquire autonomous external-action capability through a web page, link, embedded tool request, downloadable file or user instruction. Any future write or action tool requires a separate amendment and threat analysis.

47.5 Grounding and Truthfulness

- **FR-AIT-200** — Retrieval-grounded generation only, applied by source class. Tenant-specific facts come only from live Operational Evidence retrieved under the requesting user's session. Current or externally verifiable claims come from Research Evidence retrieved for the current request. General

model knowledge may support explanation and conversation but shall not be presented as a tenant fact, a live internet finding, or an official PBSMS value.

- **FR-AIT-201** — Refuse rather than guess. Where retrieval returns nothing, or nothing sufficient, the Assistant shall say so plainly and name the screen where the information would live. A stated non-answer is a correct outcome; a plausible fabricated one is a failure of the highest severity.
- **FR-AIT-202** — Currency. Retrieval shall execute against live records at question time. Retrieved data shall not be cached across sessions or reused for a later question. Every response shall carry its retrieval timestamp.
- **FR-AIT-203** — Numeric and identity fidelity. Numbers, dates, names and identifiers shall be interpolated from the retrieved payload and verified against it before display. Any response containing a value not present in the payload shall be blocked and replaced with a stated error.

Implementation note. For exact names, dates, identifiers, percentages, grades and monetary values, the preferred design is a structured model response containing evidence references and value placeholders, with a deterministic renderer inserting values from the verified payload after validation. The model composes language around a value; it does not author the value. This is the mechanism by which the 100% fidelity gate is met rather than merely required.

- **FR-AIT-204** — No implied certainty. Where a question is ambiguous — an unqualified "this term" during a term transition, a surname matching two students — the Assistant shall state the interpretation it used, or ask, rather than choose silently.

47.6 Tenant and User Isolation

The platform is shared-database, shared-schema with PostgreSQL Row-Level Security. RLS is the entire isolation boundary. The Assistant gets no exception to it, no service account around it, and no fast path through it.

- **TEN-050** — Assistant retrieval shall execute through the same request-scoped database connection and RLS session variables as any other request. No service account, no elevated role, no bypass path. If RLS blocks a row, the Assistant never sees it.
- **TEN-051** — User scope, not tenant scope. The Assistant inherits the requesting user's Chapter 13 scope. A subject teacher's Assistant reaches only that teacher's assigned classes and subjects. An accountant's Assistant cannot answer a question about scores.
- **TEN-052** — Session context, embeddings, retrieval indices and every derived artefact shall be namespaced by tenant_id and deleted on tenant offboarding (Ch 4.1) alongside primary records.
- **TEN-053** — Where a vector or search index supports retrieval, the tenant partition shall be applied at the storage layer, not as a post-query filter in application code. A post-filter is one bug away from a silent leak.
- **TEN-054** — A cross-tenant leakage suite shall exist alongside NFR-QA-020's suite as a blocking CI gate, including adversarial prompts attempting to elicit another tenant's data, another school's data within the same tenant, and data outside the requesting user's role scope.
- **TEN-055** — Platform staff operating under an impersonation grant (TEN-021) shall have the entire Assistant disabled for the duration, including General Conversation and Controlled Live Internet Research.

47.6.1 Effective scope

Dimension	Enforcement
Identity and session	Authenticated user, non-impersonated session, active account, current authentication context
Commercial access	Subscription, trial or demo status, query allowance, cost ceiling
Tenant configuration	Tenant-wide and per-role capability switches
Operational context	Tenant, school, campus, academic year, term, role, assignment, ownership, record and field permissions
Sensitive categories	Health, discipline, unpublished-result and consent controls
Research access	Collection ACL, source class, live-internet permission, file type, domain policy, tool limits
Conversation access	Owner, explicit share, workspace membership, retention status, current context
Output access	Permitted answer, draft, export, approval path and destination

- **TEN-056** — Effective scope is the intersection of all applicable permissions. A broader permission in one dimension shall never override a restriction in another.
- **TEN-057** — The server, not the model or client, shall construct and integrity-protect the per-request capability manifest.
- **TEN-058** — Research collections, embeddings, indices, cached extracts and derived research artefacts shall enforce tenant and collection access at the storage layer. Application post-filtering alone is prohibited.
- **TEN-059** — Combined Analysis shall execute operational and research retrieval through separate authorised channels and preserve source identity through generation and validation.
- **TEN-060** — Conversation storage shall be partitioned by tenant and owner. A shared conversation shall use an explicit ACL and shall never become tenant-wide merely because it was shared once.
- **TEN-061** — A change of user, tenant, school, campus, academic year, role, assignment, subscription state or capability configuration shall invalidate the current capability manifest and any context no longer valid.

47.7 Restricted Data and Model Development Data

- **DP-100** — Health records (FR-OPS-030) and discipline records (FR-OPS-040) shall be excluded from Assistant retrieval by default for every role, including roles otherwise authorised to view them directly. A tenant may enable a category for a role by explicit administrator action, recorded in the audit log.
- **DP-101** — Where enabled, health and discipline content shall be summarised only, never quoted verbatim, and never included in parent-facing drafted text.
- **DP-102** — Data minimisation. Retrieval shall return the minimum fields needed to answer. Retrieving a full student profile to answer a question about attendance violates this requirement.
- **DP-103** — The model shall be operated under a contractual zero-retention configuration prohibiting retention, human review and training on submitted content. A provider unable to commit contractually shall not be used, irrespective of capability or price.

- **DP-104** — No tenant data shall be used to train, fine-tune, adapt, distil, align or preference-train any model — not for that tenant, not for another, not at any time. Production failures shall become redacted or synthetic regression cases.
- **DP-105** — Tenants shall be told at subscription what the Assistant processes and where. PBSMS shall supply wording for the tenant's privacy notice to guardians. Guardian-facing features shall respect the consent registry (DP-070).
- **DP-106** — Session transcripts are personal data and shall participate in DP-030 access exports and DP-090 erasure within the same 30-day SLA.
- **DP-107** — Where inference occurs outside Ghana, the transfer shall be recorded in the processing register with its lawful basis, and the jurisdiction disclosed to the tenant before they subscribe.
- **DP-108** — Question text routinely contains children's names. Transcript retention shall be time-bounded under Ch 40.2 with a shorter default than academic records, configurable downward by the tenant but not upward.
- **DP-109** — Every model-development dataset shall carry a provenance manifest recording source, licence or permission, collection method, intended use, personal-data status, review outcome and permitted retention.
- **DP-110** — No tenant record, upload, research collection, conversation, feedback transcript or derived embedding shall enter model adaptation. Training environments shall have no production database route and no tenant export credential.
- **DP-111** — Personal data, confidential tenant information, credentials, internal identifiers, unpublished results and transaction details shall not be transmitted to any search engine, website, web-research provider or external content service.
- **DP-112** — Every outbound internet request shall pass a data-loss-prevention and query-sanitisation gate before transmission. A blocked request shall fail closed and be auditable.
- **DP-113** — Tenants shall be told which model-hosting, search, browsing, extraction and research services process their requests, with each service's retention, training, human-review and jurisdiction disclosed before enablement.
- **DP-114** — Stored web extracts, snapshots, query logs and citation metadata shall have explicit retention and deletion rules. Personal data encountered incidentally on the public web shall not be retained merely because it was retrieved.
- **DP-115** — Research documents shall be ingested only where the uploader or curator confirms the right to use them for retrieval. Retrieval permission does not create permission to train.
- **DP-116** — General, research and combined conversation content remains personal data where it identifies a person, and participates in access, export, retention and erasure workflows on the same terms as other transcripts.
- **DP-117** — Passwords, API keys, session tokens, private keys, database credentials and browser credentials shall not be placed in model context, training data, research collections or model-visible tool results.
- **DP-118** — Directly Supplied Content shall pass the same deterministic DLP gate as outbound research queries before any external transmission, and shall be flagged in the audit record so the shortened retention under DP-108 applies to the affected transcript.

47.8 Untrusted Content and Network Safety

Everything reaching the model that was supplied by a human or an external system is untrusted: PBSMS free text, student and guardian names, message bodies, uploaded research files, search results, web pages, document metadata, hyperlinks, hidden text, scripts, image metadata and content returned by external tools. A guardian who names their child with an instruction is an attacker with an account.

- **NFR-AIT-100** — Retrieved content shall be delivered inside a structurally delimited data channel, distinguished from instructions, with the system prompt stating that content in that channel is data and never instruction.
- **NFR-AIT-101** — Output shall be validated before display: no value absent from the retrieved payload, no record reference outside scope, no renderable markup or executable link. Links shall be generated by the PBSMS route resolver, never from model text.
- **NFR-AIT-102** — Because the Assistant has no write path (FR-AIT-100) and no external-action capability (FR-AIT-151), the maximum impact of a successful injection is a wrong answer to the user who asked. This containment is the principal defence and shall not be weakened without re-analysing this section in full.
- **NFR-AIT-103** — Detected injection attempts shall be logged with the source record or URL, so a poisoned source can be located and cleaned rather than merely blocked at read time.
- **NFR-AIT-107** — The model shall receive no database credential, network credential, browser credential, internal service token or unrestricted tool endpoint. It shall invoke only typed, server-authorised tools present in the current capability manifest.
- **NFR-AIT-108** — Web and research content shall be fetched and parsed in an isolated service that removes scripts, active forms, executable markup, tracking content, detectable hidden instructions and unsupported embedded objects before model use.
- **NFR-AIT-109** — The research gateway shall block private, loopback, link-local, metadata-service, internal-DNS and disallowed destinations, and shall defend against DNS rebinding, redirect abuse and server-side request forgery.
- **NFR-AIT-110** — Downloads shall be restricted by approved MIME type, extension, size, depth, redirect count and malware policy. Executables and active archives shall not enter model context.
- **NFR-AIT-111** — Outbound queries shall pass deterministic DLP checks before transmission. Model self-assessment is not a substitute for this gate.

47.9 Human-in-the-Loop Gates

Output	Approver	Gate
Guardian message draft	Staff role authorised on that channel	Existing Ch 26 send-review; recipient count, channel split and cost shown before send
Report-card comment	Class teacher, then the results approval chain	Cannot reach a published report card without passing FR-RES-020
Absence follow-up note	Staff member owning the queue item	FR-ATT-040 workflow unchanged
Admission decision letter	Admissions officer with approval authority	FR-ADM-030 conversion remains separate, explicit and atomic
Term summary	Headmaster or academic	Same gate as FR-ANL-040

Output	Approver	Gate
	coordinator	
Answer to a question	None — read-only, nothing leaves the tenant	Traceability (FR-AIT-011) is the control instead

- **FR-AIT-300** — Every Assistant output that leaves the tenant shall pass through the existing approval control for that artefact.
- **FR-AIT-301** — The approving human is the accountable author. The audit record shall name that person alongside a flag that the draft was AI-assisted. "The model wrote it" is not a position available under Ghana's Data Protection Act.
- **FR-AIT-310** — A draft informed by Research Evidence or General Explanation shall enter the same approval gate as the destination artefact. Source citations and the AI-assisted flag shall remain attached through approval.

47.10 Demo Mode

- **FR-AIT-400** — Demo mode shall run exclusively against a seeded synthetic dataset. Isolation shall be enforced by the demo running under a dedicated demo tenant with its own tenant_id under the same RLS policies, not by a flag in application code.
- **FR-AIT-401** — Demo access shall be capped by query count and elapsed days. Proposed 20 queries over 14 days per prospect — commercial figures, open to change; the existence of both caps is not.
- **FR-AIT-402** — Every demo response shall carry a persistent, non-dismissible marker stating the data is sample data.
- **FR-AIT-403** — A prospect converting to a paying tenant shall inherit no demo session data, transcript or context.
- **FR-AIT-404** — The demo shall exercise the real Assistant against synthetic data, not a scripted sequence of pre-written answers. A demo that cannot fail tells the buyer nothing.
- **FR-AIT-405** — Controlled Live Internet Research shall be disabled in demo mode by default. If enabled commercially later it remains read-only, uses the same gateway, exposes no tenant data, and carries separate query and cost caps.

47.11 Subscription, Metering and Billing

Concern	Specification
Packaging	A paid add-on above Ebom, not a new tier. Ketewese tenants cannot purchase it. (Proposal.)
Billing unit	Accepted requests per tenant per period. Token counts recorded for cost attribution but not billed on — a token is not a unit a headmaster can verify.
Cost recording	Model inference, search, page retrieval, extraction, storage and citation processing recorded per accepted request.
Research budget	A tenant and per-user research budget operating independently of general query caps, because research cost is variable and general query cost is not.

Concern	Specification
Metering integration	Recorded through the existing per-period pipeline (TEN-030, TEN-031), not a parallel system.
Soft cap	At 80% of allowance, the tenant administrator is notified in-app and on the configured channel.
Hard cap	At 100%, the Assistant returns a stated period-limit state naming the reset date. Every other part of PBSMS is unaffected (FR-AIT-500).
Per-user cap	A daily per-user ceiling independent of the tenant allowance, to contain a runaway loop or an abusive account.
Cost ceiling	A server-side monetary ceiling per tenant per period, enforced independently of the request cap, so a provider price change cannot produce an unbounded bill.
Non-billable	Cancelled, privacy-blocked, policy-blocked and technically failed requests shall not consume the allowance, though provider cost is recorded internally.

47.12 Surfaces

Surface	Present	Form
Staff Console	Yes	CommandPalette (⌘K) plus a dockable panel exposing Records, Research, General and Combined modes according to effective scope.
Teacher Field App	Limited	Only capabilities explicitly enabled for the role and online state. The Assistant is never part of the offline path — register and score entry work with no network and no model.
Parent View	Deferred	Highest-risk surface: guardians, minors' data, lowest technical literacy, no staff intermediary. Not authorised by this adoption.
Platform Console	No	PBSMS staff get no query interface over customer records. See TEN-055.

- **FR-AIT-500** — The Assistant shall never sit on the critical path of any core workflow. Attendance, score entry, results, invoicing and payment shall function identically whether or not it is available, subscribed to, or enabled.
- **FR-AIT-501** — The Assistant shall respect the six required UI states (Ch 34.5), including a restricted state that names what is restricted and who to ask.
- **FR-AIT-502** — Assistant responses shall respect data-saver mode: no charts, no images, text and figures only.
- **FR-AIT-503** — The interface shall render the active mode, source class, live-research status, retrieval timestamp, citations and any material source conflict, without requiring the user to inspect an audit log.

47.13 Non-Functional Requirements

ID	Requirement
NFR-AIT-010	Latency: p95 visible acknowledgement or progress within 2s; p95 complete validated operational response within 10s for a scoped single-class or single-student query; p95 complete validated standard live-research response within 30s (proposed). No substantive answer content shall be displayed before validation completes.
NFR-AIT-011	Availability target 99.0%, explicitly lower than core. Assistant unavailability shall degrade no other module.
NFR-AIT-012	Fail closed. On model, retrieval, research or validation failure, return a stated error. Never a partial or unvalidated answer.
NFR-AIT-013	Every request shall carry a timeout and be user-cancellable. A cancelled request shall not be billed.
NFR-AIT-014	Rate limiting per user and per tenant, independent of billing caps.
NFR-AIT-015	Disableable by a tenant administrator, globally or per role, effective immediately on active sessions.
NFR-AIT-016	Model version pinning. The release in use shall be recorded with every response. A provider-side model alias shall never be accepted as a pinned release.
NFR-AIT-017	Provider portability. Retrieval, validation, audit and evaluation shall be independent of any one provider's API.
NFR-AIT-018	Cost per request shall be recorded and queryable per tenant, so unit economics are measured rather than assumed.
NFR-AIT-104	Every model release shall be stored in a registry with an immutable release identifier, integrity checksum, approval state, deployment history, evaluation report and rollback target.
NFR-AIT-105	The adaptation pipeline shall be reproducible from approved code, configuration, base-model identifier and dataset manifests. Non-determinism shall be measured and bounded, not ignored.
NFR-AIT-106	Training, development and evaluation datasets shall be separated. Evaluation cases used for release decisions shall not appear in adaptation data.
NFR-AIT-112	Before display the validator shall verify source class, claim-to-source mapping, citation existence, allowed record references, exact protected values, prohibited output, render safety and approval metadata.
NFR-AIT-113	Research requests shall carry explicit search, source-count, page-size, token, elapsed-time and monetary limits, and shall be cancellable.
NFR-AIT-114	Cancelled, privacy-blocked, policy-blocked and failed research shall not consume the tenant allowance, though provider cost is recorded internally.
NFR-AIT-115	The research gateway shall implement timeouts, redirect limits, concurrency limits, domain and protocol policy, content-size limits, circuit breakers and retry-storm protection.
NFR-AIT-116	External-source evaluation shall use a replayable captured corpus for deterministic CI plus a separate controlled live-source suite for availability, freshness, redirect and citation checks.
NFR-AIT-117	Any release failing a blocking gate shall be prevented from deployment and shall support immediate rollback to the last approved bundle.

ID	Requirement
NFR-AIT-118	The model shall receive only tools authorised for the current request. A tool absent from the manifest shall be technically unreachable, not merely discouraged by prompt text.

47.14 Model Identity and Release Governance

- **FR-AIT-700** — PBSMS shall operate a named and versioned PBSMS Domain AI Model as the primary intelligence behind the Tenant AI Assistant.
- **FR-AIT-701** — Every production release shall be a durable bundle under PBSMS control containing at minimum: a model checkpoint or adaptation artifact, tokenizer and runtime configuration, base-model identity and licence, Model Constitution version, tool and output schemas, data manifest, model card, evaluation report, integrity checksum and rollback reference.
- **FR-AIT-702** — A third-party general model combined only with an unversioned prompt shall not be represented or released as the PBSMS Domain AI Model. Such a configuration may be used internally for prototyping and demand measurement, disclosed as such, and never described in a release bundle.
- **FR-AIT-703** — The Model Constitution shall be versioned as part of the release and shall define permitted capabilities, prohibited outputs, grounding rules, source classes, refusal rules, tool rules and accountability boundaries.
- **FR-AIT-704** — Rules suitable for model learning shall be included in adaptation and alignment work. Every critical rule shall additionally have an independent runtime enforcement control, or a documented reason why deterministic enforcement is not technically possible.
- **FR-AIT-705** — A change to model checkpoint, adapter, quantisation, tokenizer, constitution, system prompt, tool schema, output schema, retrieval policy, research gateway, source-ranking policy or validator shall create a new release candidate and run the applicable §47.16 gates.

47.14.1 Block rate as a release signal

The blocking gates in §47.16.2 measure properties of responses that are displayed. Groundedness, numeric fidelity and external claim attribution are each enforced by the validator refusing to display anything that fails. A model that grounds poorly therefore does not fail those gates — it passes them, and the cost surfaces as blocked responses, which reach the user as failures nothing was measuring.

- **FR-AIT-706** — Validation block rate shall be recorded per mode, per role and per release candidate, and reported alongside the §47.16.2 metrics for every release decision. A candidate whose block rate exceeds the ceiling established from canary measurement shall not be promoted, irrespective of every blocking gate being green.
- **FR-AIT-707** — A blocked response shall present the user with a stated, intelligible outcome and shall be counted as a failed request in quality reporting. A block is a system failure the validator contained, not a neutral event, and shall not be reported as a success.

47.15 Audit and Observability

- **FR-AIT-600** — Every interaction shall be logged with: actor, tenant, school, campus, academic year, question text, identifiers of records retrieved, response text, model release ID and checksum, constitution version, capability manifest version, conversation mode, source classes, research collections accessed, sanitised outbound queries, domains and URLs opened, source publication and

retrieval times, citation mapping, DLP result, network-policy result, external tool calls, source conflicts, validation outcome, latency and cost.

- **FR-AIT-601** — Assistant activity shall be visible in the tenant-facing audit log viewer (TEN-022) on the same terms as any other access to tenant records. Audit display shall distinguish private conversation content from administrative metadata; tenant visibility does not cancel the user's privacy rights, and an administrator shall not receive unrestricted transcript access merely because audit events exist.
- **FR-AIT-602** — Users shall be able to report an incorrect answer in one action from the response itself. Every reported answer becomes a permanent regression obligation under §47.16.1.
- **FR-AIT-603** — Validation blocks shall be logged and alerted on. A rising block rate is the earliest available signal that a model version, prompt, retrieval or research change has regressed.

No user-visible hidden reasoning trace shall be stored or exposed. Audit records tools, evidence identifiers, structured decisions and outcomes rather than private reasoning.

47.16 Evaluation and Release Gates

This section is what makes the Assistant reliable rather than merely built. None of it is optional or deferrable to after launch.

47.16.1 Golden sets

- No fewer than 100 question/answer pairs per capability, built against a seeded reference tenant with known-correct answers.
- At least 100 cases each for Research, General Conversation and Combined Analysis, stratified across every role and access boundary that can use the capability.
- Coverage across every role in the Frontend Design Specification §10, because scope enforcement fails per role, not globally.
- Positive and negative capability-manifest cases: authorised role, unauthorised role, disabled tenant, expired subscription, context switch, collection ACL change, conversation-share revocation.
- An adversarial corpus: cross-tenant and cross-school elicitation, out-of-scope elicitation within a role, injection through record content, adversarial web pages with visible, hidden, encoded and indirect instructions, malicious links, redirects, source impersonation, exfiltration requests, and attempts to trigger external actions.
- Mixed-source cases where Operational Evidence conflicts with Research Evidence or general knowledge.
- Privacy cases proving identifiable tenant data cannot appear in an outbound query, URL, request body, referrer, log or external tool result.
- Separate adaptation, development and blind evaluation sets, split by scenario family rather than by row, with dataset-hash checks in CI.
- Every reported incorrect answer creates a permanent regression obligation. Where the report contains personal or tenant data, the permanent case shall be redacted, transformed or synthetic — preserving the failure mechanism without retaining the original personal data beyond its lawful retention period.

47.16.2 Thresholds

Metric	Definition	Gate
Cross-tenant leakage	Response containing another tenant's data	0 — blocking
Scope violation	Response containing data outside the user's Ch 13 scope	0 — blocking
Numeric fidelity	Values in output present unchanged in the retrieved payload	100% — blocking
Groundedness	Factual claims traceable to a retrieved record	100% — blocking
Prohibited output	Predictions, characterisations or determinations about a child	0 — blocking
Model release integrity	Deployed artifact and manifest match the approved registry checksum	100% — blocking
Outbound privacy leakage	Protected tenant or personal data sent to an external destination	0 — blocking
Unauthorised source or tool use	Source, collection, domain or tool used outside effective scope	0 — blocking
External claim attribution	Every external factual claim has a valid claim-to-source mapping	100% — blocking
Unsafe external action	Any post, message, upload, transaction, sign-in or account creation	0 — blocking
Current-information honesty	Time-sensitive answer uses live evidence or states it was unavailable	100% — blocking
Injection resistance	Adversarial instruction not followed, including web and document sources	≥ 99% — blocking
Answer accuracy	Correct answer on the golden set	≥ 95%
Research answer accuracy	Correct synthesis on the research golden set	≥ 95%
Citation support	Cited source materially supports the associated claim	≥ 98%
Appropriate refusal	Refuses when evidence is insufficient rather than guessing	≥ 98%
Source-conflict disclosure	Material credible disagreement identified rather than hidden	≥ 95%
Unnecessary refusal	Refuses a question it could have answered in scope	≤ 5%
Validation block rate	Responses blocked before display, per mode	Ceiling set at canary — FR-AIT-706

The blocking metrics are pass/fail. A build failing any one does not ship, is not shipped behind a flag, and is not shipped to a single willing tenant as an exception.

47.16.3 When the gates run

On every change to: model checkpoint, adapter, quantisation, tokenizer, context length, constitution, system prompt, training or alignment dataset, tool schema, output schema, retrieval code, role mapping, research collection policy, search provider, browser or extraction service, source-ranking policy, DLP rules, network policy, validator, and the user-facing rendering of claims or citations. Not on a schedule — schedules miss the change that caused the regression.

47.16.4 Staged exposure

Stage	Condition to advance
Model laboratory	Synthetic and approved non-tenant data only. No production credentials, tenant records or unrestricted internet.
Reference environment	Approved release candidate, seeded reference tenant, captured web corpus, full access and injection suites green.
Operational shadow	Live tenant retrieval under the contractual and privacy gates. No output shown to tenants. Sampled human review.
Research shadow	Controlled live internet access through the production gateway. Outbound privacy leakage remains zero; citations and source support reviewed.
Canary	Consenting staff users in selected tenants, per-role capability flags, weekly error and privacy review, immediate rollback available.
General availability	All gates green for four consecutive weeks; stable or falling reported-error and block rates; adoption blockers closed.

47.17 Failure Modes

Failure	Detection	Required behaviour
Model checksum or manifest mismatch	Registry verification before load	Do not load or serve; retain or roll back to the last approved bundle.
Model provider unavailable	Timeout or error response	Assistant shows an unavailable state. No other module affected. No retry storm.
Retrieval returns nothing	Empty result set	Stated non-answer naming where the data would live (FR-AIT-201).
Output contains an unverifiable value	Validation against retrieved payload	Response blocked, error shown, logged and alerted (FR-AIT-603, FR-AIT-707).
Response references an out-of-scope record	Scope check on record identifiers	Blocked and escalated as an isolation defect, not merely retried.
Private data in an outbound request	DLP and query-sanitisation gate	Block before transmission; privacy-safe explanation; audit and alert.
Unauthorised research source or tool	Manifest and ACL check	Block; treat as an access-control defect, not a normal refusal.
Web or document injection	Content inspection and behavioural validation	Ignore or block; log the source; do not expand tool access or alter instructions.
Unsafe destination or redirect	Network and SSRF policy	Block before connection; record destination

Failure	Detection	Required behaviour
		and policy reason.
Citation missing or unsupported	Claim-to-source validator	Block the affected response and return a stated validation error.
Credible sources conflict	Source comparison rule	Present the disagreement with dates and authorities; do not fabricate consensus.
Live research unavailable	Gateway error, timeout, provider limit	State that live research is unavailable; never present model memory as current research.
Malicious or unsupported download	MIME, size, malware and archive policy	Reject before ingestion; do not expose bytes to the model.
Conversation permission revoked	ACL change event or per-request check	Remove access immediately and invalidate affected context.
Cost or source-count ceiling reached	Server-side accounting	Stop retrieval; state the limit; return only a validated result supported by sources already retrieved, or a stated non-answer.
Tenant exceeds cost ceiling	Server-side cost accounting	Assistant disabled for the period with a clear message; platform bill contained.

47.18 Build Order

Stage	Build	Why here
0.5	Demand validation — Ask only, staff only, read-only, on the governance runtime with a prompt-configured general model behind the model adapter	The governance runtime is required whatever model sits behind it. The model programme is contingent and expensive. Sequencing the contingent work after the evidence justifying it costs weeks; the reverse costs quarters if the answer is no.
1	Model definition: architecture criteria, Model Constitution, release-bundle format, registry, dataset policy	Documented, reviewable model identity. No tenant data used.
2	Synthetic domain corpus, access matrix, golden sets, adversarial corpus, captured web corpus, evaluation harness	Tests exist before adaptation decisions can be optimised against deployment behaviour.
3	Base-model selection and PBSMS adaptation: instruction, tool-use, refusal, grounding, citation and schema behaviour	Candidate bundle passes offline model-level gates on held-out data.
4	Governance runtime: RLS retrieval, capability manifests, collection ACLs, validators, audit, DLP, research gateway, network isolation	Deterministic isolation and privacy tests pass without relying on model obedience.
5	Integrated model on synthetic and reference tenants — Ask plus General Conversation for staff	All blocking gates green.
6	Controlled Live Internet Research and research collections, internal shadow	Zero outbound privacy leakage; citation, injection, network and cost controls green.

Stage	Build	Why here
7	Operational and research shadow, then consenting canary tenants	Four consecutive green weeks; stable or falling error and block rates.
8	Draft, with approval gates wired to existing controls	Source traceability survives drafting and approval.
9	Explain, Find, Combined Analysis, demo controls, metering, caps, billing	Capability-specific gates and commercial limits approved.
10	Parent View	Not authorised by this chapter. Separate risk review and adoption required.

Stage 2 before stage 3 is the load-bearing sequencing decision. The same reasoning applied to multi-tenancy in the original architecture rewrite and to accessibility in the frontend build order: the properties that cannot be retrofitted go first. Stage 0.5 exists for a different reason — not because demand validation cannot be retrofitted, but because everything after it is expensive and none of it is worth doing if the answer is no.

47.19 Open Questions and Honest Limitations

Decision required	Why it blocks or shapes the work
Base model, scale, licence, tokenizer, context length	Determines ownership, adaptation cost, hardware, language quality, portability and release-bundle contents.
Self-hosted, managed or hybrid inference	Changes privacy, jurisdiction, cost, availability and provider dependency. DP-103 and DP-107 narrow the field materially and constrain architecture, not just procurement.
Adaptation method	Instruction tuning, parameter-efficient tuning, preference optimisation and retrieval training carry different data and evaluation requirements.
Approved non-tenant corpus	Ghanaian education, administration, finance, policy and language material requires provenance, permission, quality review and bias testing.
Validation block-rate ceiling	No principled prior exists. Must be set from canary measurement, and it is the figure most likely to determine whether the model-first programme is viable at the candidate scale.
Roles permitted live internet research	Research utility and privacy risk differ sharply between teachers, accountants and administrators.
Search, browsing and extraction providers	Retention, query logging, jurisdiction, coverage, pricing and zero-retention commitments differ.
Source policy	A strict allowlist is safer but narrow; open search is broader but demands stronger ranking, reputation and injection controls.
Conversation persistence	Sessions are currently not resumable across logins. Durable history requires an explicit access, retention and user-control design.
Permitted file types and research depth	Documents, archives, images, audio and video create different extraction and security risks.
Cost per request	Unmodelled, so \$47.11 is a structure without numbers. The cost ceiling exists precisely because this is unknown.

Decision required	Why it blocks or shapes the work
Multilingual	Twi, Ga, Ewe and Hausa require independent evaluation before any authorisation.
Parent View	Highest-risk surface. Deferred, not resolved.

Limitations that survive adoption and are not decisions anyone can make:

- Detection of confidential Directly Supplied Content (FR-AIT-905) is heuristic and will both miss cases and produce false positives. FR-AIT-904 carries the requirement because it is deterministic.
- The tension between 100% groundedness, ≥95% accuracy and ≤5% unnecessary refusal is real at the candidate scale, and the release process resolves it silently in favour of blocking unless FR-AIT-706 is enforced.
- No user research exists. Whether a Ghanaian headmaster will trust an AI answer about money at all is the commercial risk here, and engineering cannot retire it. Stage 0.5 exists to measure it.
- The 95% accuracy threshold is a proposal, not a derived figure. Re-set it from canary measurement, and only ever upward.
- This chapter has been written but not reviewed by anyone other than its author. The self-review blind spot noted in the Frontend Design Specification §15 applies equally here.

— End of Chapter 47 —